Off-Grid Solar Sizing Report – Rural clinic (example)

Prepared by: - Date: - Status: OK

Summary (bill of materials)

Solar panels	**1 × 550 W = 550 W array**
Battery bank	**4 × 12.8 V / 100 Ah units (1S4P) = 12.8 V, 400 Ah, 5,120 Wh**
Inverter	**500 W continuous (limited by surge)**
MPPT charge controller	**60 A at 12.8 V**
DC fuses	**Battery-inverter: 63 A; controller-battery: 63 A**
DC cables (copper)	**Battery-inverter: 16 mm² (2 m); controller-battery: 16 mm² (3 m)**
AC output breaker	**6 A MCB**

Calculation steps

1. Daily energy	AC 1,560 Wh ÷ 0.92 + DC 0 Wh = 1,695.7 Wh/day (continuous AC load 215 W)
2. System voltage	Auto: 215 W continuous and 1,696 Wh/day → 12 V; bank = 1 × 12.8 V = 12.8 V
3. PV array	1,695.7 Wh ÷ (5 h × 0.77) = 440.4 W → 1 × 550 W = 550 W; est. yield 2,118 Wh/day
4. Battery bank	1,695.7 Wh × 2 days ÷ (0.8 × 0.95) = 4,462.2 Wh → 3.49 strings → 4 in parallel; 1S4P = 400 Ah, usable 4,096 Wh
5. Inverter	max(215 W × 1.25 = 268.8 W; surge 615 W ÷ 2 = 307.5 W) → 500 W (limited by surge)
6. MPPT controller	550 W ÷ 12.8 V = 42.97 A × 1.25 = 53.7 A → 60 A
7. Battery-inverter cable	500 W ÷ (12.8 V × 0.92) = 42.46 A; fuse 53.1 A → 63 A; drop over 2 m needs 7.74 mm²; ampacity >= fuse → 16 mm² (66 A,
8. Controller-battery cable	42.97 A; fuse 53.7 A → 63 A; drop over 3 m needs 11.75 mm²; ampacity >= fuse → 16 mm² (66 A, drop 2.20 %)
9. AC breaker	500 W ÷ 230 V × 1.25 = 2.72 A → 6 A MCB

Warnings

Count	None.

Assumptions used

Site	Peak sun hours 5 h/day; derate 0.77; autonomy 2 days; max DC voltage drop 3 %
Battery	Unit 12.8 V / 100 Ah; depth of discharge 0.8; round-trip efficiency 0.95
Inverter / controller	Efficiency 0.92; safety factor 1.25; surge multiple 2; controller safety factor 1.25; diversity 1; AC 230 V

Preliminary design aid, not a stamped engineering design. Check all results against local electrical codes, manufacturer datasheets and site conditions with a qualified person. Cable ampacities are indicative (single-core PVC copper, free air, ~30 C).